

5. Information was collected about accidents on the *Seapron* bypass. It was found that the number of accidents per month could be modelled by a Poisson distribution with mean 2.5

Following some work on the bypass, the numbers of accidents during a series of 3-month periods were recorded. The data were used to test whether or not there was a change in the mean number of accidents per month.

- (a) Stating your hypotheses clearly and using a 5% level of significance, find the critical region for this test. You should state the probability in each tail. (5)
- (b) State $P(\text{Type I error})$ using this test. (1)

Data from the series of 3-month periods are recorded for 2 years.

- (c) Find the probability that at least 2 of these 3-month periods give a significant result. (3)

Given that the number of accidents per month on the bypass, after the work is completed, is actually 2.1 per month,

- (d) find $P(\text{Type II error})$ for the test in part (a) (3)



1. The number of customers entering Jeff's supermarket each morning follows a Poisson distribution.

Past information shows that customers enter at an average rate of 2 every 5 minutes.

Using this information,

- (a) (i) find the probability that exactly 26 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning, (2)
- (ii) find the probability that at least 21 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning. (2)

A rival supermarket is opened nearby. Following its opening, the number of customers entering Jeff's supermarket over a randomly selected 40-minute period is found to be 10

- (b) Test, at the 5% significance level, whether or not there is evidence of a decrease in the rate of customers entering Jeff's supermarket. State your hypotheses clearly. (4)

A further randomly selected 20-minute period is observed and the hypothesis test is repeated. Given that the true rate of customers entering Jeff's supermarket is now 1 every 5 minutes,

- (c) calculate the probability of a Type II error. (5)



- (f) Carry out the test using a 5% level of significance. (4)

5. Asha, Davinda and Jerry each have a bag containing a large number of counters, some of which are white and the rest are red.
Each person draws counters from their bag one at a time, notes the colour of the counter and returns it to their bag.

The probability of Asha getting a red counter on any one draw is 0.07

- (a) Find the probability that Asha will draw at least 3 white counters before a red counter is drawn. (2)
- (b) Find the probability that Asha gets a red counter for the second time on her 9th draw. (2)

The probability of Davinda getting a red counter on any one draw is p .
Davinda draws counters until she gets n red counters. The random variable D is the number of counters Davinda draws.

Given that the mean and the standard deviation of D are 4400 and 660 respectively,

- (c) find the value of p . (4)

Jerry believes that his bag contains a smaller proportion of red counters than Asha's bag. To test his belief, Jerry draws counters from his bag until he gets a red counter. Jerry defines the random variable J to be the number of counters drawn up to and including the first red counter.

- (d) Stating your hypotheses clearly and using a 10% level of significance, find the critical region for this test. (5)

Jerry gets a red counter for the first time on his 34th draw.

- (e) Giving a reason for your answer, state whether or not there is evidence that Jerry's bag contains a smaller proportion of red counters than Asha's bag. (2)

Given that the probability of Jerry getting a red counter on any one draw is 0.011

- (f) show that the power of the test is 0.702 to 3 significant figures. (3)



- (2)

She records the number of calls for help received on each of these days.

- (4)

(4)

Given that over the next n days $P(C = 0) < 0.001$

- (3)

- (1)

[illegible]

- (4)

2. The number of errors made by a secretary is modelled by a Poisson distribution with a mean of 2.4 per 100 words.

A 100-word piece of work completed by the secretary is selected at random.

- (a) Find the probability that

(i) there are exactly 3 errors,

(ii) there are fewer than 2 errors.

(2)

After a long holiday, a randomly selected piece of work containing 250 words completed by the secretary is examined to see if the rate of errors has changed.

- (b) Stating your hypotheses clearly, and using a 5% level of significance, find the critical region for a suitable test.

(4)

- (c) Find $P(\text{Type I error})$ for the test in part (b)

(1)

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Answer ALL questions. Write your answers in the spaces provided.

1. Bacteria are randomly distributed in a river at a rate of 5 per litre of water. A new factory opens and a scientist claims it is polluting the river with bacteria. He takes a sample of 0.5 litres of water from the river near the factory and finds that it contains 7 bacteria. Stating your hypotheses clearly test, at the 5% level of significance, whether there is evidence that the level of pollution has increased.

(5)

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7. Sam and Tessa are testing a spinner to see if the probability, p , of it landing on red is less than $\frac{1}{5}$. They both use a 10% significance level.

Sam decides to spin the spinner 20 times and record the number of times it lands on red.

- (a) Find the critical region for Sam's test. (2)

- (b) Write down the size of Sam's test. (1)

Tessa decides to spin the spinner until it lands on red and she records the number of spins.

- (c) Find the critical region for Tessa's test. (6)

- (d) Find the size of Tessa's test. (1)

- (e) (i) Show that the power function for Sam's test is given by

$$(1 - p)^{19}(1 + 19p)$$

- (ii) Find the power function for Tessa's test. (4)

- (f) With reference to parts (b), (d) and (e), state, giving your reasons, whether you would recommend Sam's test or Tessa's test when $p = 0.15$ (4)